JÖNKÖPING UNIVERSITY
School of Engineering

IMPACT OF WEATHER CONDITIONS ON SWEDEN'S ENERGY PRODUCTION

Submitted by

Group number: 9

Anaina Biju Sulfitha

Ardra Suresh

Ashiq Abdul Rahim

School of Engineering, Jonkoping University

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I. INTRODUCTION

Renewable energy such as wind and solar, hydroelectric power is extremely important for producing electricity in Sweden. The thing about these energy sources that's really, interesting is that the weather can affect how much electricity you can make. For instance, windmills require wind to generate electricity. The panels need weather to function. Hydropower plants must have rain to produce electricity.

The purpose of the work is to give a better view of how different kinds of weather affects the production on electricity in Sweden. It's extremely important to understand such matters when one considers the very reality that Sweden is one of the "renewable saturated" countries! The data provides the opportunity to learn more about climate-dependent variability and how transforming weather affects electricity generation.

In this work, we integrate weather data from SMHI with energy production data by SCB. The study is for the period between May 2024 and 2025 September data from four weather stations are used Göteborg, Malmö, Helsingborg and Gävle. These stations were chosen to sample diverse regions and climates in Sweden. The analysis has been performed through the KNIME Analytics Platform. And after applying data cleaning, aggregation, visualization and clustering we can see how weather patterns correlate with temporal shifts in energy production.

II. DATA PROCESSING

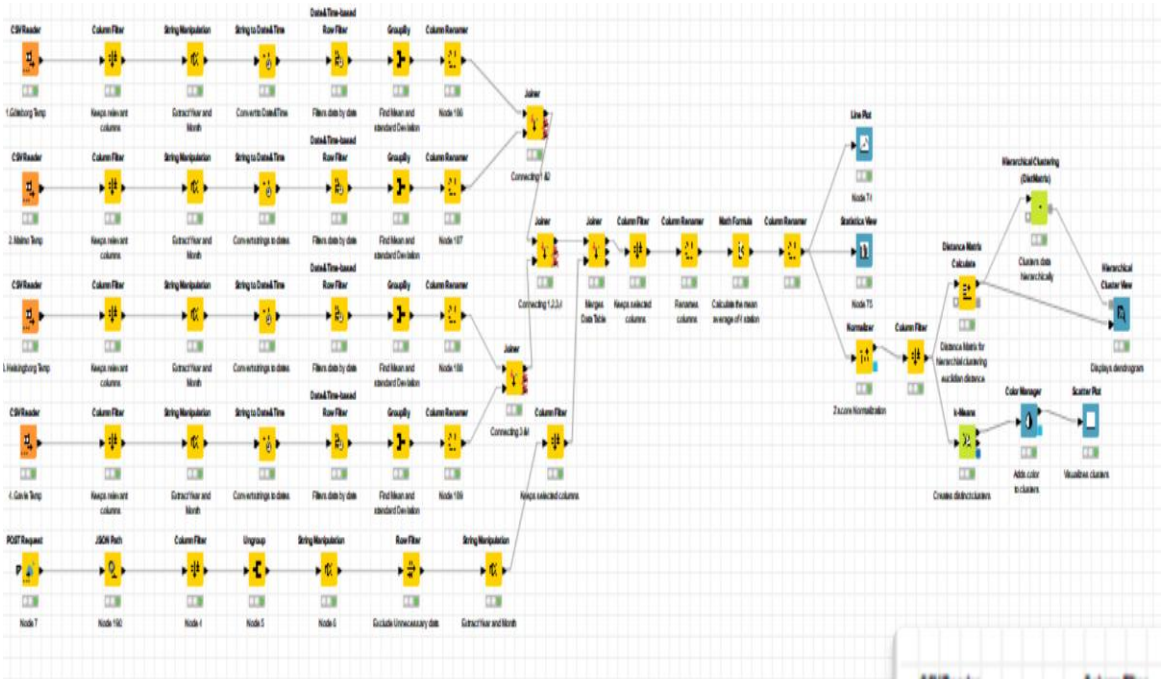
The workflow was carried out with the KNIME Analytics platform visual data workflows software, following step by step. The flow starts with retrieving data from two public APIs, SMHI and SCB, for weather observations and energy production statistics.

Three weather variables were chosen from the SMHI API average temperature, average wind speed, and average humidity. This was selected due

to importance in renewable energy generation. Data on weather comes from four stations: Göteborg, Malmö, Helingsborg and Gävle.

The data on weather was preprocessed and aggregated in the form of monthly. Minimum, maximum, median values and standard deviations were also calculated beside average values to describe the variability of transferred data. The information from all four stations were then merged into one table. We sourced energy production for the model from the SCB API, which provides renewable energy production on monthly time steps. This data was processed to be consistent with the corresponding period for weather data. A join was employed, in which each row in the joined dataset represents one month with both weather and energy variables.

Some visualizations were developed to investigate the correlation between weather and energy generation. Hierarchical clustering was then performed to cluster months with similar characteristics.



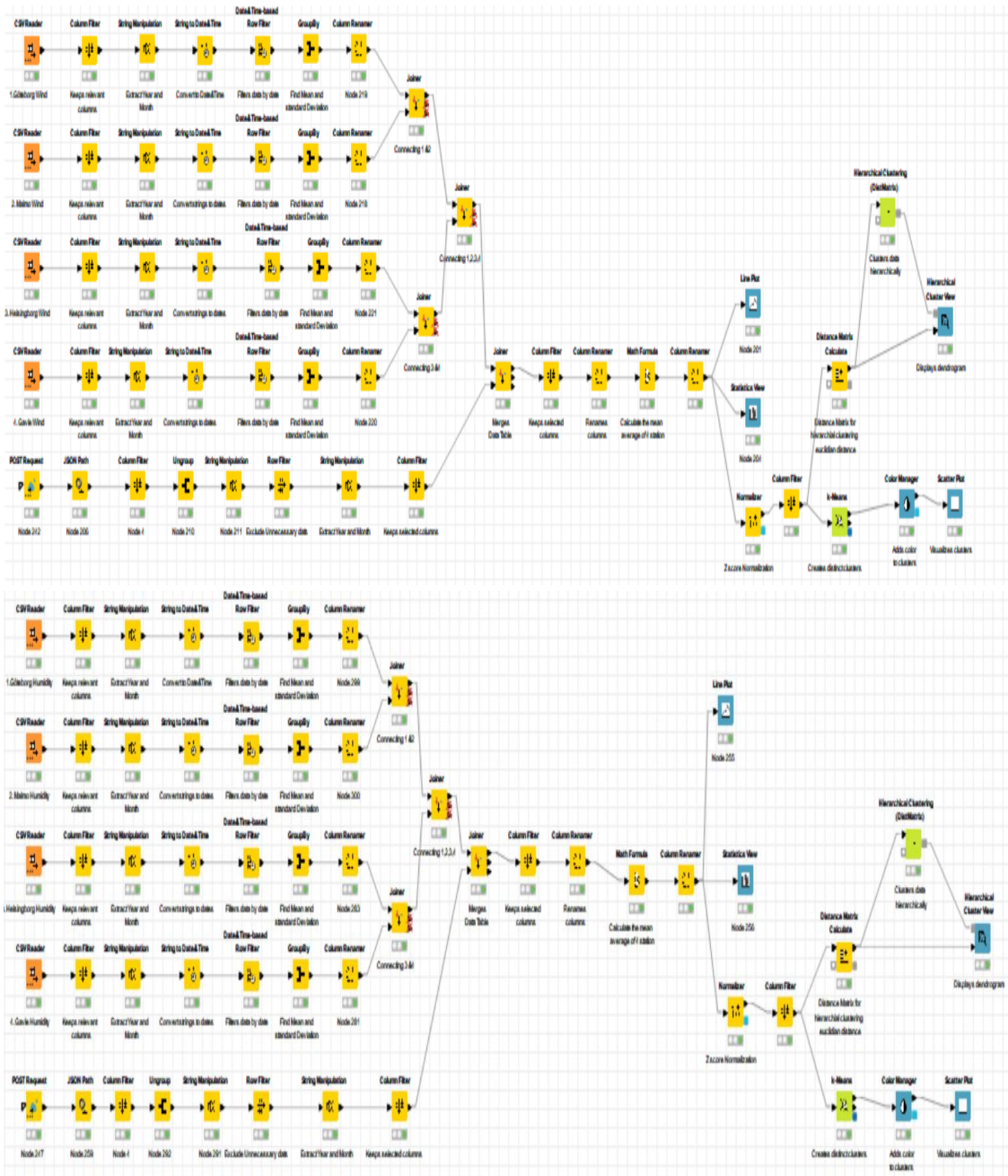


Figure 1: Knime workflow with separate pipelines for each weather parameter

III. CLASSIFICATION AND CLUSTERING APPROACH

For studying patterns in the data, a classification-oriented procedure based on hierarchical clustering was used. As the dataset is not labelled, clustering with

similar clusters was used to cluster months of similar weather and energy characteristics together.

We employed the identical KNIME workflow structure for all weather parameters (temperature, wind speed and humidity) to maintain consistency. First, only the relevant super merged variables (i.e. monthly weather values from all four stations and energy production) were chosen for clustering.
The data was normalized by Z-score normalization before clustering. step as the variables are measured on disparate scale. This is to prevent any one variable from having undue affect over the clustering.

Subsequently, a distance matrix was derived from the centroids based on Euclidean distance. The hierarchical clustering algorithm was applied on the distance matrix.

An agglomerative hierarchical clustering was conducted, in which each month starts as its own cluster and similar months are grown together. These results are presented as a dendrogram that shows how clusters form at different distance levels three clusters were chosen as a balance between detail and interpretability. The resulting cluster assignments were re-added to the data and employed for visualization in subsequent interpretation.

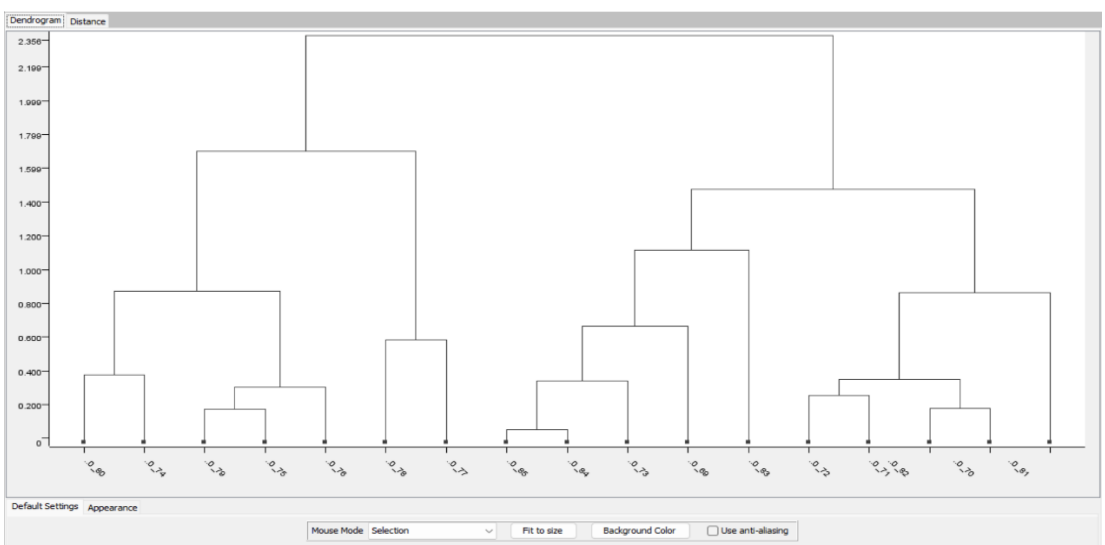


Figure 2: Hierarchical clustering dendrogram

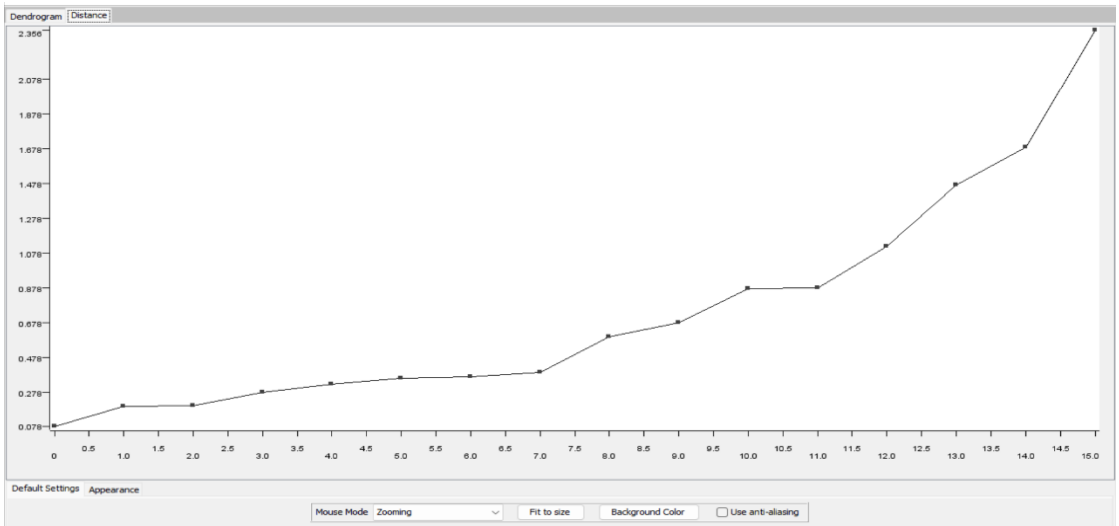


Figure 3: Distance plot used to select the number of clusters

IV. RESULTS AND DISCUSSION

4.1 Humidity patterns across stations

The result suggests relatively high humidity level in the coastal locations such as Göteborg and Helsingborg compared to the inland stations. These patterns are present in analyses of hydropower production as high humidity typically is associated with increased precipitation and water availability.

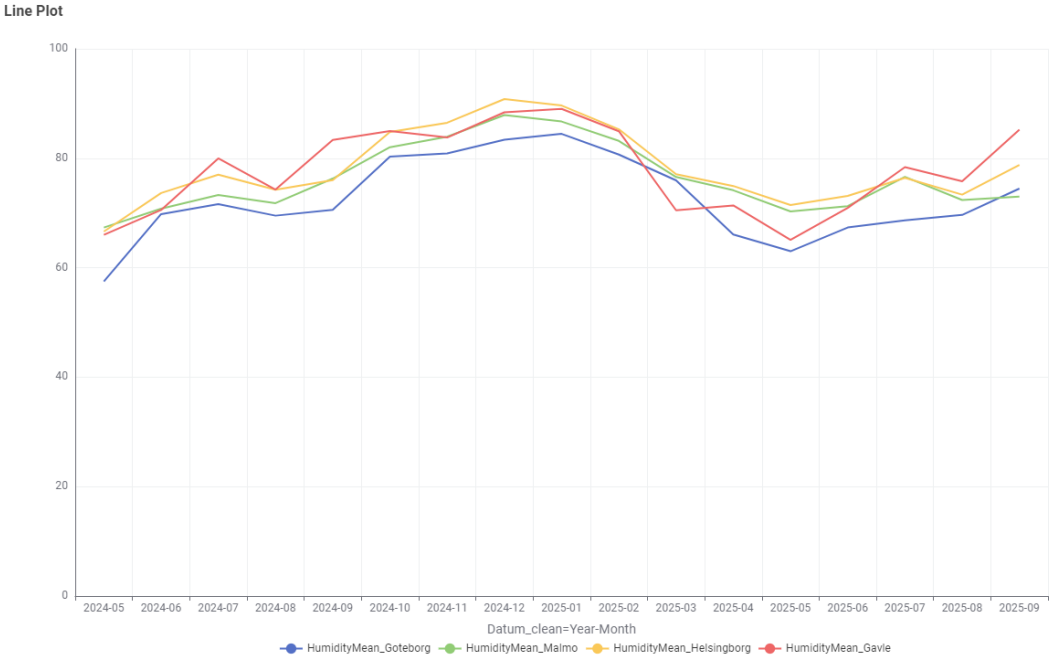


Figure 4: Line plot showing monthly average humidity for all four stations

4.2 Wind Speed

The Line Plot reveals that average wind speeds generally register a higher value during winter than in summer. Winds tend to be stronger in the coastal areas, especially at Malmö and Helsingborg. These findings can partly explain why wind turbine generation is higher in fall and winter. Results indicate that wind speed is an important parameter on the performance of wind power.

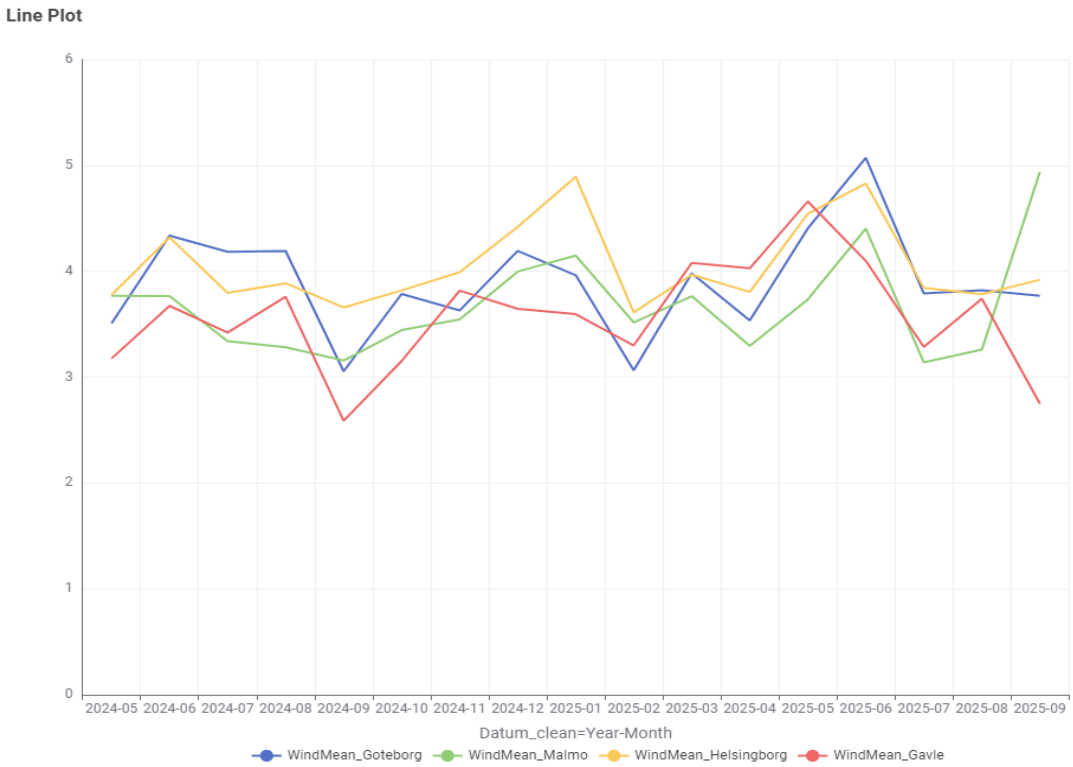


Figure 5: Line plot showing monthly average wind speed for all four stations

4.3 Temperature and hydropower relationship

The scatter diagram of mean temperature and hydropower production indicates a negative relationship. Higher levels of hydropower production are generally more associated with lower temperatures and less with higher temperatures. This effect may be explained by higher precipitation and snow-melting in colder periods.

The clusters are color-labelled to indicate a closeness in season. Hotter with less hydropower production are in one cluster, and cooler and wetter with

higher hydro output are in another. This illustrates how temperature impacts hydropower production indirectly via the seasonal climate.

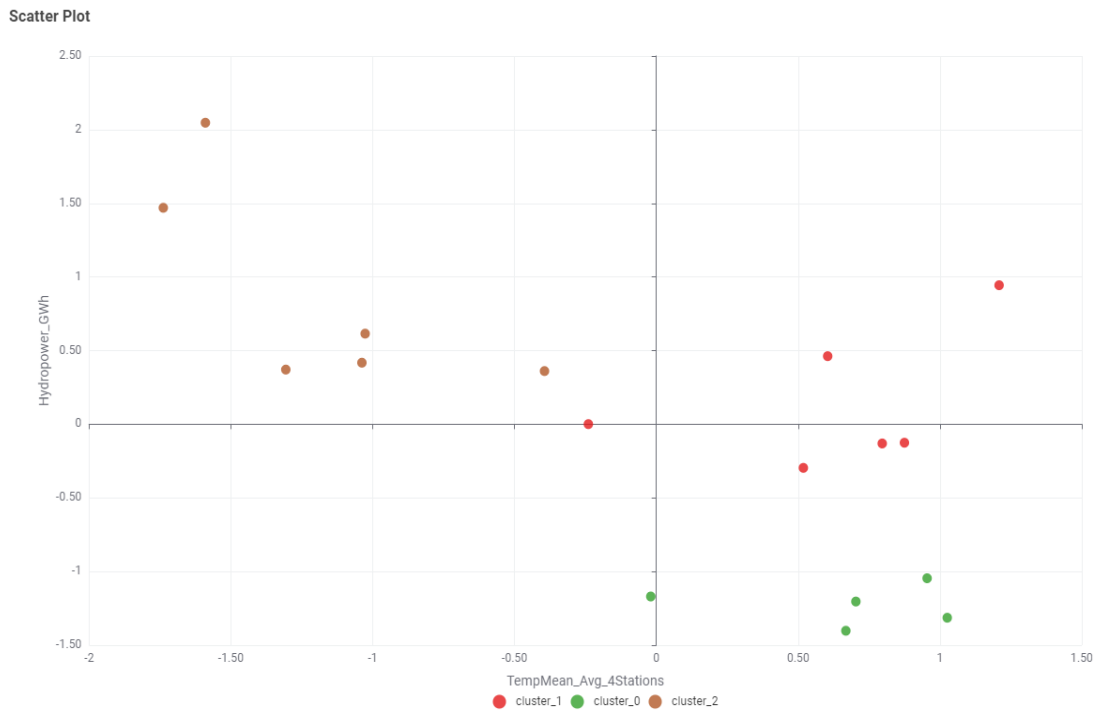


Figure 6 – Scatter plot of temperature versus hydropower production, colored by cluster

V. CONCLUSION

The present article demonstrates that the weather has a significant effect on the generation of RE in Sweden. We used SMHI weather data and combined it with energy production SCB data to find strong seasonal trends and interesting correlations between the proposed proposals of weather conditions and produced efficient power.

Wind was overall stronger in the fall and winter periods, which is the same period when you tend to get better wind energy production. Temperature showed a well-defined seasonal pattern, and hydropower generation increased with colder and wetter months. These results were confirmed by the hierarchical clustering result, where months were divided into some seasons of similar characteristics.

In terms of learning, on the assignment side it was useful to work with APIs and clean/aggregate data from different sources, create visualizations and apply some clustering methods in KNIME. It also underscored that technological workflows are simply not enough and there is a need to clearly interpret analytical results.

In conclusion, the results draw attention to how weather fluctuations are essential in renewable energy generation and illustrate how prospective data-driven analysis can contribute to improved energy planning and policy making in Sweden for future energy planning.